📊 Power BI Sales & Marketing Analytics Project

📌 Overview

This project is an end-to-end Business Intelligence solution built in Power BI, focused on analyzing e-commerce performance across sales, customer behavior, product performance, and marketing campaigns.

The goal of the project is to transform raw transactional and campaign data into actionable insights through data modeling, transformation, and interactive visualizations.

🎯 Business Questions Answered

This dashboard answers key business questions such as:

- 💰 How is revenue evolving over time?
- 📦 Which product categories drive the most revenue and transactions?
- 👥 How do different customer segments behave?
- 📈 What is the average order value (AOV) across channels and products?
- 📣 Which marketing campaigns and channels are most effective?
- ⏱️ Do shorter or longer campaigns perform better?
- 🔄 What is the relationship between transaction volume and revenue?

🧱 Data Modeling

A **star schema** was implemented to ensure scalability and performance:

- **Fact Tables:**
  - transaction_fact
  - events_fact
- **Dimension Tables:**
  - product_dim
  - campaigns_dim
  - customer_dim
  - date_dim (custom-created)

🔹 Key Modeling Features

- Created a dedicated **Date Table** to enable proper time intelligence and relationships
- Built a separate **Measures Table** to organize all DAX calculations

- Implemented relationships between fact and dimension tables
- Handled missing keys using **“Unknown” members** to maintain referential integrity

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🔄 Data Transformation (Power Query)

Data was cleaned and prepared using Power Query:

- Removed or filtered invalid records (e.g., NULL dates, invalid transactions)
- Created calculated columns such as:
  - Campaign duration (in days)
  - Campaign duration groups (Short, Medium, Long)
- Converted fields (e.g., seconds → minutes)
- Standardized categorical values (e.g., handling "Unknown" categories)
- Ensured consistent data types across all tables

📐 DAX Measures

Core business metrics were created using DAX:

- **Total Revenue**
- **Total Transactions**
- **Unique Customers**
- **Average Order Value (AOV)**
- **Transactions per Customer**
- **Revenue per Campaign**
- Additional filtered measures using:
  - CALCULATE
  - DISTINCTCOUNT
  - DIVIDE

A **valid transaction flag** was used to ensure all KPIs are based on clean and reliable data.

📊 Dashboard Pages & Analysis

🏠 Overview Dashboard

- High-level KPIs
- Revenue trends over time

- Revenue by category and channel
- Interactive slicers for filtering

📦 **Product Analysis**

- Revenue contribution by category
- AOV comparison (premium vs non-premium)
- Top-performing brands
- Category-level performance trends

📣 **Campaign Analysis**

- Revenue by marketing channel and objective
- Campaign performance by duration groups
- Target segment effectiveness
- Campaign efficiency insights (e.g., revenue per duration)

💳 **Transaction Analysis**

- Revenue vs transaction trends
- Volume vs value analysis (AOV vs transactions)
- Customer behavior insights
- Distribution of transaction sizes
- Scatter analysis of revenue vs transactions

⚙️ **Advanced Features**

- 🔄 Interactive slicers and filters
- 🎯 Drill-down capabilities for hierarchical analysis
- 🧭 Page navigation buttons
- 🔁 Bookmark-based “Reset Filters” functionality
- 📊 Dual-axis charts for multi-metric comparison
- 📌 Custom sorting (e.g., Month, Weekday)

🧠 **Key Insights**

- Revenue drivers vary significantly between categories and channels
- Some segments generate high transaction volume but low value (and vice versa)
- Campaign duration impacts efficiency, with shorter campaigns often performing better
- Customer behavior differs across segments and channels